

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

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Class : SYIT	Batch : 1
Date of Assignment : 22/7/26	Date/Time of Submission : 22/7/26

Questions:

2a. Write a python code to create nested tuples.

```
a = [(4, 5, 'Cprog'), (7, 8, 6)]

student = ("Shubh", 11, ("BSCIT", "Semester III"), 90)

print("Name =", student[0])
print("Roll No =", student[1])
print("Department =", student[2][0])
print("Exam =", student[2][1])
print("Percentage =", student[-1])
print("S009")
```

```
= RESTART: C:/Users/mvluu
PDS2.py
Name = Shubh
Roll No = 11
Department = BSCIT
Exam = Semester III
Percentage = 90
S009
```

2b. Write a python code to sort the nested tuple using sorted () function.

```
data = [(1, 4), (3, 2), (5, 1)]

sorted_data = sorted(data, key=lambda x: x[1])

print(sorted_data)
print("S009")
```

```
= RESTART: C:/Users/mvluc
PDS2.py
[(5, 1), (3, 2), (1, 4)]
S009
|
```

2c. Write a python code to copy or clone list.

```
LIST1 = [10, 20, 30, 40, 50]

LIST2 = LIST1.copy()

print("List NUMBER 1:", LIST1)
print("List NUMBER 2:", LIST1)
print("S009")

= RESTART: C:/Users/mvluc/AppData/Lo
PDS2.py
List NUMBER 1: [10, 20, 30, 40, 50]
List NUMBER 2: [10, 20, 30, 40, 50]
S009
|
```

2d. Write a python code to check the immutability property of python tuples.

```
t= ("Shubh", "me")
t[1]="i"
print("S009")
```

```
= RESTART: C:/Users/mvluc/AppData/Local/Programs/Python/Python314/PDS2.py
Traceback (most recent call last):
  File "C:/Users/mvluc/AppData/Local/Programs/Python/Python314/PDS2.py", line 2, in <module>
    t[1]="i"
TypeError: 'tuple' object does not support item assignment
|
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80